

Project Title:
**Conversion of post-combustion gas from cement manufacturing
to syngas by electrochemical CO₂ capture and reduction**

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August monthly report

1. Tasks list for August

- Catalyst modification:
 - a. Axial coordination of NiSAC catalyst with triazole derivatives
 - b. Physical blending of Ni0.5SAC with MOFs
 - CO₂RR performance measurement of modified catalysts under diluted CO₂ and
- ### 2. Experimental Procedure

Preparation of Ni0.5SAC@PANI and cathode electrode

Briefly, Ni0.5SAC (150mg) and PANI (15, 50, and 75mg) were mixed in 15ml of IPA to make a dispersion solution. Subsequently, the dispersion solution was sonicated for 2h, followed by overnight stirring at a room temperature. Afterwards, the dispersed Ni0.5SAC@PANI was filtered out, followed by several washing with DI. The solid residue was then overnight dried at 80°C to finally obtain Ni0.5SAC@PANI powder. 150mg of Ni0.5SAC@PANI was added into a 10ml of IPA, followed by adding 381μL of XC-2 binder and sonicating for another hour to make an ink solution. The ink solution was then sprayed onto Sigracet 39BB carbon paper to obtain a mass loading of approx. 0.18 ± 0.1 mg/cm².

Preparation of Ni0.5SAC@MOFs cathode electrode via spray coating method

Briefly, Ni0.5SAC (150mg) and 15mg of different types of MOFs (CALF20, CALF20-dtz-atz, and CALF20-3MeI) were sonically blended in 10ml of IPA and 381μL of XC-2 binder to make a dispersion solution. The ink solution was then sprayed onto Sigracet 39BB carbon paper to obtain a mass loading of approx. 0.18 ± 0.1 mg/cm².

Axial coordination modification of Ni0.5SAC with triazole derivatives

The synthesis of axial coordinated Ni0.5SAC was done by a previous method.¹ Briefly, 150mg of Ni0.5SAC was added into a 50ml round-bottom flask. Afterwards, the large excess of axial ligands (mole ratio of ligand and Ni-SAC of 10:1) dissolved in 10ml DMF was added into the round bottom flask and stirred at 75°C for 24h. After cooling down to room temperature, the precipitate was filtered and washed several times with ethanol. Finally, the precipitate was overnight dried at 80°C to obtain ax-Ni0.5SAC-tz, atz, and dtz powder.

Triazole derivatives: 1,2,4-triazole (tz), 3-diamino-1,2,4-triazole (atz), and 3,5-diamino-1,2,4-triazole (dtz)

Electrochemical CO₂RR measurements

A 5 cm² MEA electrolyzer was used to evaluate the CO₂RR performance under modulated operating conditions. The Ni foam, bipolar membrane (BPM), and different cathode catalysts were employed as the anode, membrane, and cathode electrode, respectively. 50 ml of 1M KOH was recirculated and used throughout the experiment as the anolyte at 8 ml/min. The humidified 15%CO₂/85%N₂ gas stream was introduced into the cathodic inlet under total flow rates of 100 ml/min. Meanwhile, the cathodic outlet was directly injected into an online gas-chromatography for gas products analysis.

3. Results

Influence of physical blending of Ni0.5SAC with polyaniline (PANI) on CO₂RR performance under 15%CO₂ condition

As shown in Fig. 1, the modified NiO.5SAC@PANI exhibited similar results to bare-NiO.5SAC with a maximum FE(CO) of 73% and highly stable under a diluted 15%CO₂ at 200mA/cm² for 3h.

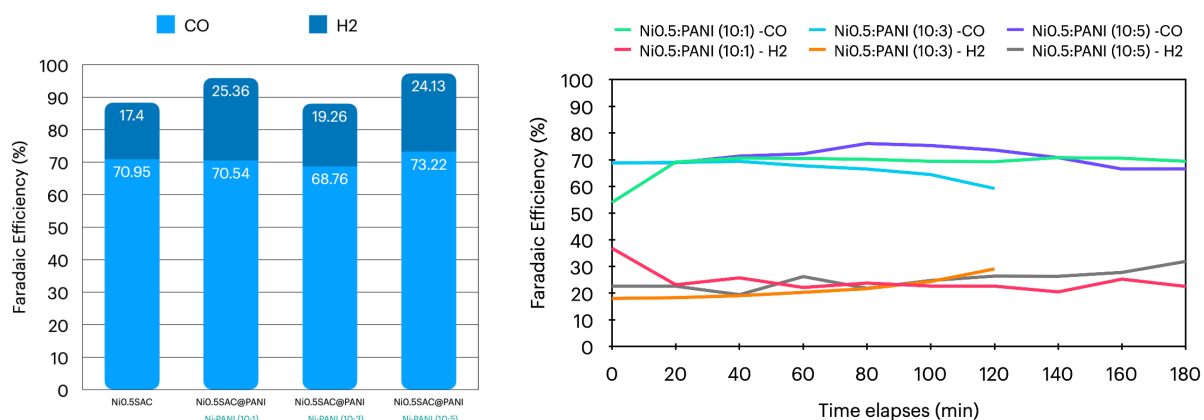


Figure 1. Faradaic efficiency of bare-NiO.5SAC and NiO.5SAC@PANI under a diluted CO₂ condition.

Influence of physical blending of NiO.5SAC and MOFs on CO₂RR performance under 15%CO₂ condition

Further studies on the physical blending of NiO.5SAC and CALF20 derivatives were conducted. As can be seen in Fig. 2, all NiO.5SAC@MOFs showed poor performance and stability compared to a bare-NiO.5SAC due to its high-water adsorption under a high humidity environment, thereby causing GDE flooding and high HER activity.

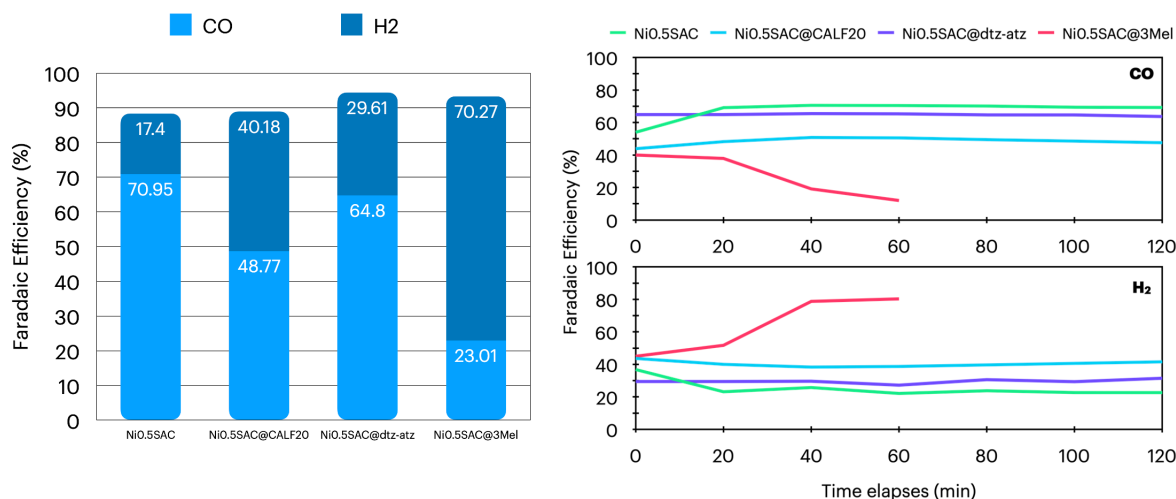


Figure 2. Faradaic efficiency of bare-NiO.5SAC and NiO.5SAC@MOFs under a diluted CO₂ condition.

Influence of axial modification of NiO.5SAC with triazole derivatives on CO₂RR under 15%CO₂ condition

Inspired by a few reports demonstrating the enhanced electrochemical CO₂ reduction reaction of axial coordination that could regulate the electronic structure and single-atom catalyst environment to be highly selective toward CO₂RR, the axial modification of Ni_{0.5}SAC with triazole derivatives was conducted.^{1–4} As shown in Fig. 3, By increasing the amine functional group of the triazole linker, the CO₂RR performance further decreased, which might be attributed to an enhance of hydrophilicity of the catalyst facilitates a competing HER activity over CO₂RR.

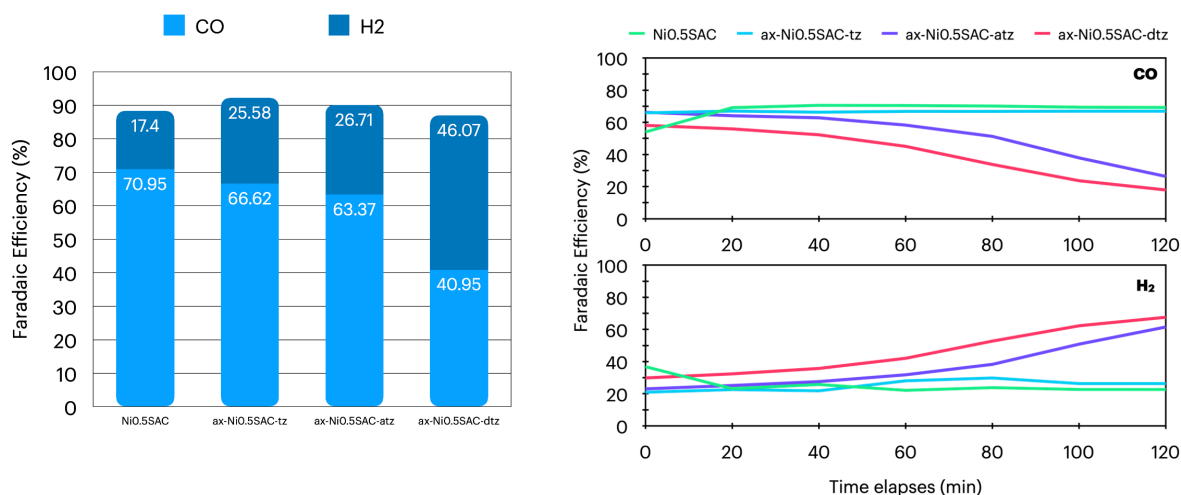


Figure 3. Faradaic efficiency of bare-Ni_{0.5}SAC and ax-Ni_{0.5}SAC under a diluted CO₂ condition.

4. Upcoming tasks

- Literature review on developing single atom catalysts.
- CO₂RR under NO_x and SO_x containing CO₂ feed.
- Studies of pulse CO₂ electrolysis under a diluted CO₂ condition.

5. References

1. Kang, H. et al. Enhancing CO₂ Reduction Efficiency on Cobalt Phthalocyanine via Axial Ligation. *ChemCatChem* 15, e202300576 (2023).
2. Ma, M. & Tang, Q. Axial coordination modification of M–N₄ single-atom catalysts to regulate the electrocatalytic CO₂ reduction reaction. *J Mater Chem C Mater* 10, 15948–15956 (2022).
3. Lv, R., Chen, S. & Wang, H. Axial Coordination Regulating Single Atoms for Enhanced CO₂ Electroreduction. *ChemElectroChem* 11, e202300857 (2024).
4. Liu, X., Liu, Y., Yang, W., Feng, X. & Wang, B. Controlled Modification of Axial Coordination for Transition-Metal Single-Atom Electrocatalyst. *Chemistry – A European Journal* 28, e202201471 (2022).